

EDUCATION

- United States citizen. Graduated from Broad Run High School with an Advanced Diploma in 2023:
- George Mason University. B.S. Computer Science -- Fairfax, Virginia || Expected graduation: May 2027.
- Relevant coursework: Essentials of Computer Science, Python, Object-Oriented Programming (Java), C and Unix, Data Structures, Data Sciences (R Programming), Statistics for Engineers, MySQL and SQL, and Computational Languages.

TECHNICAL SKILLS

- Languages: Python (Pandas, NumPy, yFinance), Java, C, R programming (Plotly, Imvar, SQL(MySQL, PostgreSQL), JavaScript (NodeJS, React.js), Brightscript (Roku).
- Tools/Platforms: Docker, AWS CodeBuild/CodeDeploy, Git, VMWare, VSCode, Eclipse, PyCharm.
- Testing: Postman, JUnit., Thunder Client for VSCode.
- Other Tools: Microsoft Office 365 suite, Google Suite, Slack, Shell Scripting, JIRA.
- Soft Skills: Critical thinking, time management, teamwork, adaptability, problem-solving, driven by initiative.

EXPERIENCE

AVCA - Office Intern - 08/2024 - 09/2024

- Engineered Python scripts to digitize 30% of the company archives and automate reporting, realizing a 15% increase in departmental productivity.
- Reduced survey processing time by 200% through automated keyword indexing; streamlined Agile operations by implementing new communication infrastructure.

Mathnasium LLC - Instructor - 09/2023 - 11/2024

- Responsible for instructing students, keeping an organized workspace, managing paperwork, rescheduling, and grading.

PROJECTS

Portfolio Performance Visualizer - Personal Project - January 2025

- Developed a Python-based analysis tool using yfinance and Pandas to fetch and process historical market data for diverse asset classes.
- Engineered interactive dashboards with Plotly to visualize portfolio distribution, cumulative returns, and volatility through dynamic charts.

CRUD API development - Personal Project - October 2025

- Developed a create, read, update, and delete API service which seamlessly interacts with a PostgreSQL database, and updates the database.
- Tested through a Docker container and with Pull and Get requests through both the Thunder Client for VSCode and through Postman.
- Used NPM to install powerful tools that automate database configuration, and created JavaScript files to communicate with PostgreSQL directly.

HackFax x Patriothacks CTF Developer, Tester, Volunteer - December 2025 - February 2026

- Participated as a developer and tester for the Capture the Flag challenges for the Hackfax event.
- Developed one easy, medium, and hard challenge, all consisting of website-based attack platforms.
- Developed a medium Joint Webattack/OSINT challenge and led/coordinated the Joint Challenge team to finish team objectives.
- Volunteered for the 3-day Hackathon/CTF event, including setting up, judging, troubleshooting, and challenge walkthroughs.

ETF Analysis Dashboard Platform - Personal Project - April 2026

- Full-stack financial tool calculating alpha, beta, and implied upside (DDM) for 34 ETFs, utilizing Plotly for dynamic multi-page visualizations.
- Integrated complex datasets via pandas-datareader to automate the calculation of the Efficient Frontier and liquidity metrics (Log Dollar Volume).
- Optimized application performance through modular code architecture (data layer vs. model layer) and proactive handling of historical data gaps.